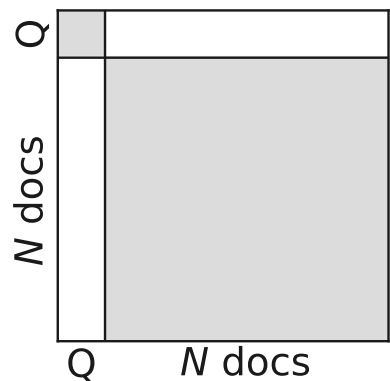
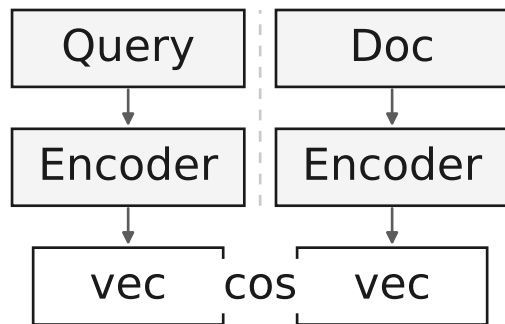


Bi-encoder

embedding pipeline

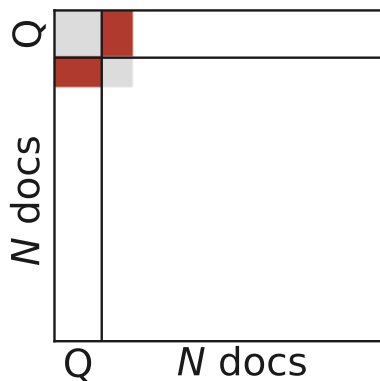
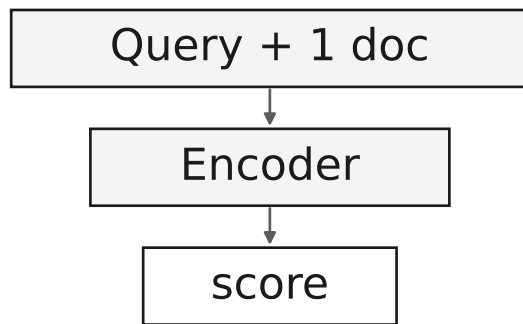


never jointly

N vector comparisons
per query

Cross-encoder

pairwise reranker

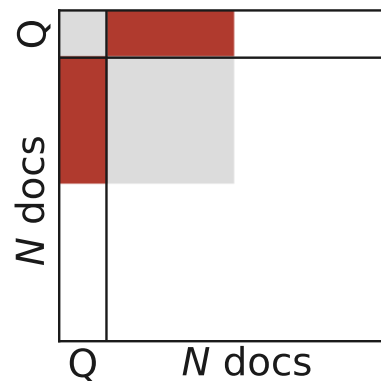
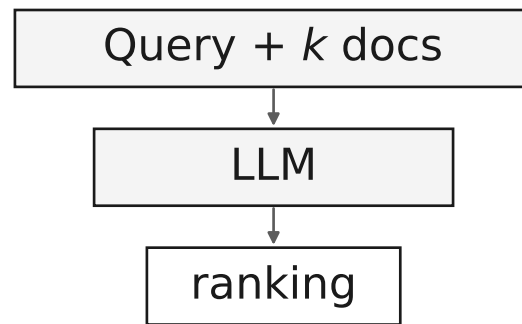


one document at a time

k pairwise passes
per query

LLM listwise

reranker over top- k

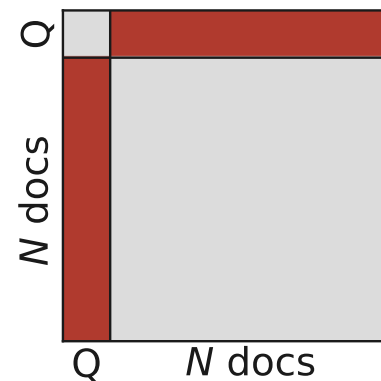
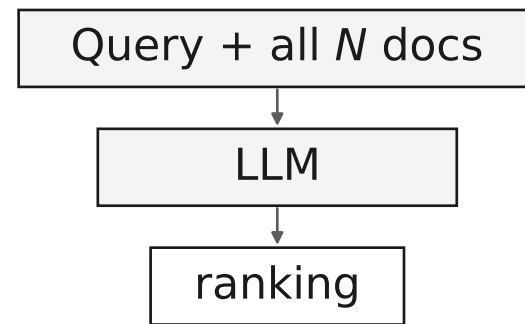


k documents at once

one pass over k docs
per query

LLM corpus-in-context

generative retrieval



the whole corpus at once

one pass over N docs
per query



self-attention



query-document



no attention